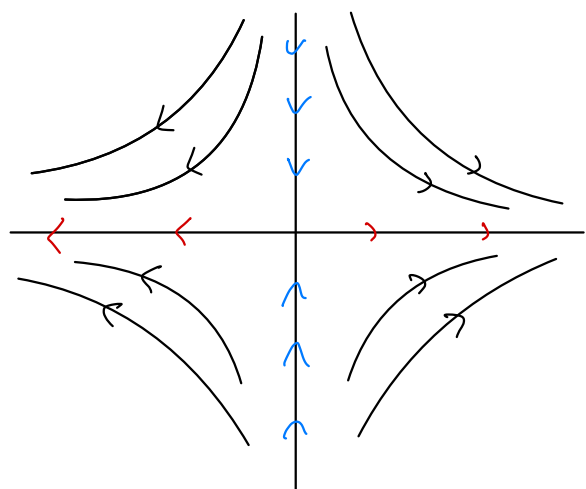


Introduction to Modern Theory of Dynamical Systems.

Katok - Hasselblatt.

- Smooth dynamics $\begin{cases} \nwarrow \text{ergodic theory} \\ \leftarrow \text{topological dynamics} \\ \nwarrow \text{Hamiltonian dynamics} \end{cases}$
- flows $\longleftrightarrow$ UF $\longleftrightarrow$ ODE.
- flows $\longleftrightarrow$ maps ($\Rightarrow$: time 1 map, $\Leftarrow$: suspension)
- topology $\longleftrightarrow$ smooth
- contraction/expanding (by linearization)
- local analysis $\Rightarrow$ global picture.
- eg) homoclinic $\Rightarrow$ Smale horseshoe
- complexity $\uparrow$

Ex. $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $(x, y) \mapsto (2x, y/2)$: hyperbolic "linear" map.



$$f^n(x, y) = (2^n x, 2^{-n} y) \\ \Rightarrow xy = \text{Const.}$$

p : homoclinic pt $\Rightarrow f^n(p)$: homoclinic.

1.5. linear flow on the torus & completely integrable systems.

$$\cdot \quad \mathbb{T}^2 = \mathbb{R}^2 / \mathbb{Z}^2, \quad \omega = (\omega_1, \omega_2)$$

$$\dot{x}_1 = \omega_1, \quad \dot{x}_2 = \omega_2$$

$$T_\omega^t(x_1, x_2) = (x_1 + \omega_1 t, x_2 + \omega_2 t) \pmod{1}$$

$\gamma = \omega_2 / \omega_1$: rational / irrational.

$$\cdot \quad \dot{x}_i = \omega_i \quad \text{on} \quad \mathbb{T}^n = \mathbb{R}^n / \mathbb{Z}^n$$

$$\Rightarrow T_\omega^t(x_1, \dots, x_n) = (x_1 + \omega_1 t, \dots, x_n + \omega_n t) \pmod{1}$$

Prop. $\{T_\omega^t\}$: minimal $\Leftrightarrow \omega_1, \dots, \omega_n$: rationally indep.
Orbit is dense.

sketch) $\cdot \quad \sum k_i \omega_i = 0 \quad (k_i \in \mathbb{Z})$

$$\Rightarrow \sin(2\pi \sum k_i x_i) : \text{inv. under } T_\omega^t \Rightarrow \text{non dense orbit}$$

$$\cdot \quad \sum k_i \omega_i \neq 0$$

$$t = k / \sum k_i \omega_i : \text{countably many}$$

$\Rightarrow$ We can choose t_0 s.t. $T_\omega^{t_0}$: minimal translation.

$$\cdot \quad \dot{x} = Ax \quad (A : 2n \times 2n, \text{ eigenvalues } \pm i\alpha_i)$$

$$\Rightarrow x(t) = e^{tA} x(0)$$

$$\begin{pmatrix} R_{t\alpha_1} & & \\ & \ddots & \\ & & R_{t\alpha_n} \end{pmatrix} \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}$$

$$\Pi_p^k = \{ |z_1| = p_1, \dots, |z_n| = p_n \} \quad (p : \text{nonnegative vector})$$

$$\dim \Pi_p^k = k = \# \text{ nonzero } p_i \text{'s}$$

$\hookrightarrow \Pi_p^k$ by translation.

$\Rightarrow \Pi_p^k$: invariant tori w.r.t. the flow

$$\dot{x}_i = \frac{\partial H}{\partial x_{i+n}} \quad \& \quad \dot{x}_{i+n} = -\frac{\partial H}{\partial x_i}$$



Ham. v.f. X_H by $\omega(-, X_H) = dH$.

Hamiltonian system is completely integrable in VCM

if $\exists$ coord. $(I, \varphi) = (I_1, \dots, I_n; \varphi_1, \dots, \varphi_n)$

(action-angle coordinate) action angle

$\omega = \sum d\varphi_i \wedge dI_i$ & H depends only on I

$$\Rightarrow X_H(I, \varphi) = (0, \dots, 0, -\partial H / \partial I_1, \dots, -\partial H / \partial I_n).$$

$\pi_n^c := \{ I_i = c_i \}$: n -torus & $\varphi_{X_H}^t$: linear on π_n^c .

$\omega(I) := (\partial H / \partial I_1, \dots, \partial H / \partial I_n)$: frequency vector.

- On the torus, $\varphi_{X_H}^t = \varphi_{X_H}^0 - t\omega(I) \pmod{1}$
- rational independence of $\omega(I) \Rightarrow$ dynamics on torus.

Thm (Arnold-Liouville)

$\{f_i, f_j\} = 0$, $\{Df_i\}$: lin. indep. on

$$M_z = \{x \in M : f_i(x) = z_i\}$$

$\Rightarrow M_z$: Lag. & inv. under $\varphi_{f_i}^t$

Moreover, if M_z : cpt & conn.

$\Rightarrow M_z \cong \mathbb{T}^n$ diffs & $\varphi_{f_i}^t$: linear on M_z .

Rmk. $\exists$ action-angle coord. near M_z .

1.8. Hyperbolic toral automorphisms.

$$L: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad (x, y) \mapsto (2x+y, x+y).$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$F_L: \mathbb{R}^2/\mathbb{Z}^2 \rightarrow \mathbb{R}^2/\mathbb{Z}^2$$

$$F_L(x, y) = L(x, y) \pmod{1}$$

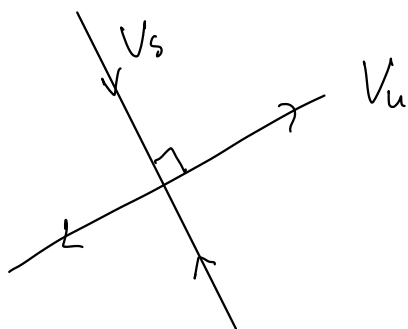
$$\lambda_u = \frac{3+\sqrt{5}}{2} > 1, \quad \lambda_s = \frac{3-\sqrt{5}}{2} < 1$$

$$(L - \lambda_u) \begin{pmatrix} 1 \\ \frac{\sqrt{5}-1}{2} \end{pmatrix} = 0$$

V_u

$$(L + \lambda_s) \begin{pmatrix} 1 \\ -\frac{\sqrt{5}+1}{2} \end{pmatrix} = 0$$

V_s



Prop. $\cdot$ $\{\text{periodic pts of } F_L\}$: dense

$\cdot F_L$: topologically transitive ($\exists x$ s.t. $\{F_L^n(x)\}_{n \in \mathbb{Z}}$: dense)

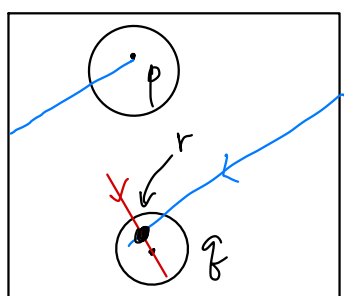
$\cdot P_n(F_L) = |\det(F_L^n - I)| = \lambda_u^n + \lambda_s^n - 2$

$\hookleftarrow$ $\#$ n -periodic pts of F_L . ($\because$ lattice pts)

sketch) $\cdot \left(\frac{s}{8}, \frac{t}{8}\right) \in \mathbb{Q}^2$: periodic pts.

$\cdot F_L^n(x, y) = (x, y) \Rightarrow \underbrace{(L^n - I)}_{\text{invertible } (\lambda_u, \lambda_s \neq 1)} \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{Z}^2$

$p \in U, q \in V$: periodic pts



$$n = (\text{period of } p) \cdot (\text{period of } q)$$

$$\begin{cases} F_L^{kn}(U) \rightarrow p \text{ as } k \rightarrow \infty \\ F_L^{kn}(U) \rightarrow q \text{ as } k \rightarrow \infty. \end{cases}$$

$$F_L^{kn}(U) \cap V \neq \emptyset \text{ for large } k.$$

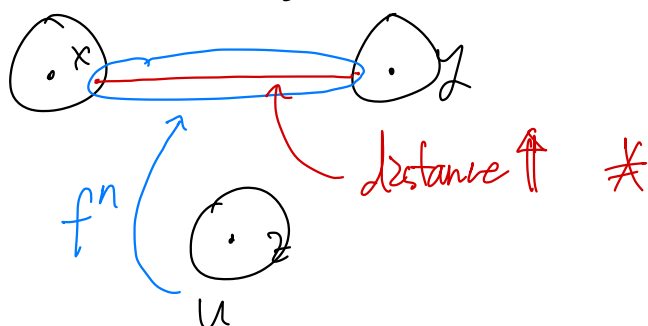
Rmk. translation / rotation / linear flow "uniform w.r.t. initial condn."

$\updownarrow$
hyperbolic automorphism "highly sensitive to initial conditions for F_L "

Def. $f: X \rightarrow X$: topologically mixing if $\forall$ open U, V
 $\exists N$ s.t. $n \geq N \Rightarrow f^n(U) \cap V \neq \emptyset$.

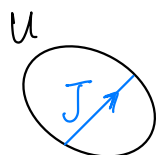
Lemma. $\forall$ metric preserving $\Rightarrow$ not topologically mixing.

sketch)



Prop. F_L : topological mixing

sketch)



$\exists N$ s.t. $F_L^N(J)$ meets every ε -ball on $\mathbb{T}^2$.

1.9. Symbolic dynamical systems.

- $\Omega_N := \{\omega = (\dots, \omega_{-1}, \omega_0, \omega_1, \dots) : \omega_i \in \{0, \dots, N-1\}\}$
- product topology of $\mathbb{Z}$ -copy of $\{0, \dots, N-1\}$ w/ discrete topology.
- $C_{\alpha_1, \dots, \alpha_k}^{n_1, \dots, n_k} := \{\omega \in \Omega_N : \omega_{n_i} = \alpha_i\}$: cylinder of rank k .
 $\uparrow$ base for the above topology. (open & closed)

$$(\lambda > 1) \quad d_\lambda(\omega, \omega') = \sum \frac{|\omega_n - \omega'_n|}{\lambda^{|n|}}$$

- Ω_N : perfect, totally disconn, cpt $\Rightarrow$ homeo to a Cantor set.

- $\sigma_N : \Omega_N \rightarrow \Omega_N$: shift transformation. (full shift).

$$\uparrow \quad (\omega_n) \mapsto (\omega'_n) \quad \omega'_n = \omega_{n+1}$$

homeo $\because$ cyl $\rightarrow$ cyl

Prop. • $\{\text{periodic pts of } \sigma_N\} = \text{dense in } \Omega_N$

$$\bullet P_n(\sigma_N) = N^n$$

- σ_N : topologically mixing.

$$\text{sketch) } \bullet \sigma_N^m \omega = \omega \iff \omega_{n+m} = \omega_n$$

$$C_{\alpha_1, \dots, \alpha_k}^{n_1, \dots, n_k}, \quad L := \sum n_i.$$

$$\omega_{n_i + kL}^\alpha = \alpha_i \quad \forall k \Rightarrow \omega^\alpha \in C_{\alpha_1, \dots, \alpha_k}^{n_1, \dots, n_k}.$$

$$\bullet \quad C_{\alpha_1, \dots, \alpha_k}^{n_1, \dots, n_k}, \quad C_{\beta_1, \dots, \beta_q}^{m_1, \dots, m_q}$$

$$L : \text{suff. large}, \quad \omega_{n_i} = \alpha_i, \quad \omega_{L+m_i} = \beta_i$$

$$\Rightarrow \omega \in \sigma_N^L(C_{\beta_1, \dots, \beta_q}^{m_1, \dots, m_q}) \cap C_{\alpha_1, \dots, \alpha_k}^{n_1, \dots, n_k}$$

Rmk. $\Omega_N^R, \mathcal{O}_N^R$: one-sided analogue.

$\pi: \Omega_2^R \rightarrow K = \text{middle third Cantor set.}$

$$(\omega_0, \omega_1, \dots) \mapsto 0.\beta(\omega_0)\beta(\omega_1)\dots$$

$$\beta(\omega_i) = 2\omega_i. \quad (\omega_i = 0 \text{ or } 1)$$

$$E_3: K \rightarrow K \quad x \mapsto 3x \pmod{1}$$

$$\Rightarrow \pi \circ \mathcal{O}_2^R = E_3 \circ \pi.$$

Def. Λ : closed, shift inv subset of Ω_N

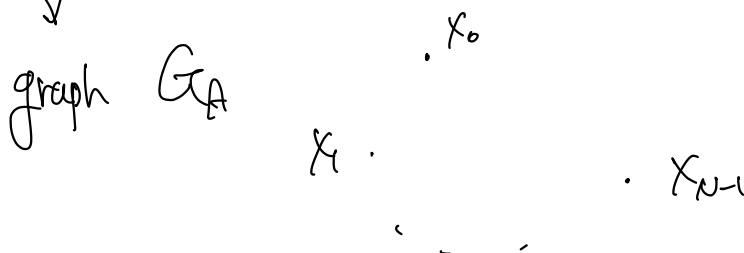
$\sigma_N|_\Lambda: \Lambda \rightarrow \Lambda$: symbolic dynamical system.

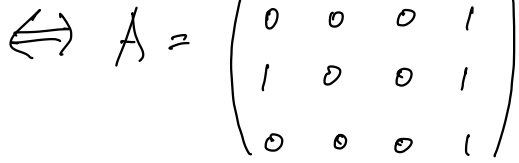
Rmk. These systems provide a rich source of (counter) ex for topological dynamical systems & ergodic theory.

0-1 matrix

$A = (a_{ij})_{i,j=0}^{N-1}$: $N \times N$ matrix & $a_{ij} = 0$ or 1 .

Def. $\mathcal{I}_A := \{\omega \in \Omega_N : a_{|n|, \omega_{|n|}} = 1 \quad \forall n \in \mathbb{Z}\}.$
 $\sigma_A := \sigma_N|_{\mathcal{I}_A}: \mathcal{I}_A \rightarrow \mathcal{I}_A$: topological Markov chain.
(subshift of finite type)




$$\Rightarrow N_{ij}^m = a_{ij}^m$$

$\therefore$ closed admissible paths $\xleftrightarrow{1-1}$ periodic pts.

Q_A : transitive if A is transitive.

(in) finite seq. of vertices in G

l.e. $a_{i, d_{i+1}} = 1$ for $i = -k, \dots, k-1$

sketch) $a_{ij}^m > 0 \Rightarrow \text{diag } A^{m+2k+1} > 0$, $\exists w_1, \dots, w_m$ s.t.

Prop. A : transitive

$$\Rightarrow \begin{cases} \mathcal{O}_A : \text{topological mixing} \\ \{\text{periodic orbits of } \mathcal{O}_A\} : \text{dense in } \Omega_A. \end{cases}$$

Def. $A: \{1, \dots, N\}^{n+1} \rightarrow \{0, 1\}$

$\Omega_A := \{\omega \in \Omega_N : A(\omega_m, \dots, \omega_{m+n}) = 1 \ \forall m \in \mathbb{Z}\}$.

$\sigma_A := \sigma_N|_{\Omega_A} : \Omega_A \rightarrow \Omega_A$: n-step topological Markov chain

Rmk. $A \sim A$

• $A, \dots, A^n \rightsquigarrow A$

• $A(i_1, \dots, i_n, j_1, \dots, j_n) = 1$

if $i_{k+1} = j_k$ & $A(i_1, \dots, i_n, j_n) = 1$.

Thm. (Perron-Frobenius).

L : non-negative $N \times N$ matrix, L^n : positive.

$\Rightarrow \exists!$ (upto scalar multi.) eigenvector e w/ positive coordz.

eigen value λ_e : simple, positive, $>$ (other λ)

Cor. A : transitive 0-1 matrix $\Rightarrow \begin{cases} P_n(\sigma_A) = \lambda_{\max}^n + \mu_n. & (\lambda_{\max} > 1) \\ |\mu_n| < C \lambda^n \text{ for some } \lambda < \lambda_{\max} \end{cases}$

sketch) $P_n(\sigma_A) = \lambda_{\max}^n + \sum_{\lambda \neq \lambda_{\max}} \lambda^n$

$Ae = \lambda_{\max} e$, λ : positive

$\Rightarrow \lambda_{\max}^m e = A^m e$, $\lambda_{\max}^m e_i = \sum a_{ij}^m e_j > e_i$.

Rmk. $\#\{\text{periodic orbits of } \sigma_A^n\} \approx \lambda_{\max}^n \Rightarrow$ "Entropy".

i.e. topological Markov chain $\Rightarrow$ complexity of periodic orbits